

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals

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- **ABSTRACT** This study evaluated an epoch-aware pipeline for EEG blink-region detection that avoided estimating detection thresholds from concatenated blink-containing and blink-free epochs. The three-stage pipeline first screened non-overlapping epochs by peak-to-peak amplitude using an autoreject-inspired cross-validation criterion, then estimated a sample-level threshold from the suspicious-epoch pool using either a mean-based or median/MAD-based estimator with a fallback for small pools, and finally extracted blink regions and mapped them to epoch-local coordinates. BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med were compared on the Internal and Cao2018 naturalistic driving-EEG corpora using event-level overlap matching and macro-averaged F_1 . At 30 s, Proposed-Med achieved a pooled macro- F_1 of 84.03%, significantly exceeding BLINKER-concat at 72.56% and MNE-annot at 60.86%. Proposed-Mean and Proposed-Med were statistically indistinguishable ($p = 1$), which indicated that the gain came from epoch screening rather than the choice of Stage-B estimator. The channel-selection ablation further showed that a single frontopolar electrode was sufficient for the strongest single-channel detection on both corpora. Finally, at Fp1 and Fp2, the two electrodes identified by the channel-selection ablation, evaluated separately, pooled macro- F_1 varied by only 0.41 and 0.50 percentage points, respectively, across epoch durations from 10 s to 120 s, and no tested duration differed from the 30 s reference after Bonferroni correction for either electrode.
- **INDEX TERMS** blink detection, electroencephalography, epoch-aware detection, naturalistic driving, threshold selection

I. INTRODUCTION

Blinking is a physiological process that involves the closure and reopening of the eyelids [1]. Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort [1]. Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research [2, 3]. Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state [2, 4]. As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance [2]. Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue [3, 5].

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography [3, 5]. Although EEG is mainly used to measure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye move-

ments generate strong electrical potentials [6, 7]. These ocular activities are especially prominent in frontal EEG electrodes due to their close proximity to the eyes [6, 8]. As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements [5, 9]. This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness [2, 10].

It is important to distinguish between two major perspectives on blink activity in the existing literature [3, 6]. In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity [6, 7]. In such cases, blink components are commonly removed during EEG preprocessing [6, 11]. However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than noise [3, 5]. From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue [5, 10]. Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches [9, 12]. Machine learning

methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape [13, 14]. Thresholding-based methods, in contrast, detect candidate blink events when the signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels [8, 15].

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals [15, 16]. The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems [16, 17]. Existing approaches range from single- or few-channel detectors that apply amplitude thresholds with rule-based decision logic to hybrid pipelines that combine thresholding as an initial detection step with additional candidate-refinement mechanisms [8, 9, 12, 14–16, 18–24]. Despite these developments, selecting an appropriate threshold remains challenging because blink amplitude varies across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events [16, 17]. This backdrop explains why simple one-value global thresholds can be attractive, yet brittle. Therefore, adaptive and robust threshold selection remains a critical consideration for reliable blink detection, particularly in fatigue-driving applications [5, 10].

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows [25]. This epoch-structured setting introduces additional challenges for threshold-based blink detection [16, 17]. When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent [11, 24]. As a result, naive application of thresholding in an epoch-structured context may risk under-detection of blinks, which undermines the reliability of fatigue assessments based on blink metrics [5, 10]. Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording [9, 24].

This paper makes several contributions to blink detection in fatigue driving research. First, we propose a double thresholding blink detection pipeline comprising epoch screening, sample level threshold estimation, and blink region extraction. To our knowledge, this is among the first frameworks to integrate these three stages explicitly to address the challenges associated with epoch structured EEG recordings. Second, we systematically compare the proposed pipeline against baseline methods, including BLINKER concat [8]

and MNE annot. To support reproducibility and facilitate further research, we also developed an automated pipeline for extracting ocular indices from EEG recordings and made it freely available as a Python toolbox called *pyblinker* (TODO). The toolbox is intended to facilitate reproducible blink analysis and encourage more consistent reporting of ocular indices in EEG based fatigue research.

The remainder of the paper is organized as follows. Section Methods and materials describes the steps and methods used to extract blinks from EEG data, as well as the datasets we used to validate the results. Section Results presents the results of running the BLINKER toolbox on a large and varied collection of EEG data. Section The BLINKER software briefly introduces the toolbox, referring readers to the more detailed online documentation and downloads. Section Visual verification of BLINKER output describes our manual video verification process and some interesting results we obtained when comparing subject behavior captured in video with blink signals captured by EEG. Section Discussion offers some concluding remarks and discusses limitations of the method.

II. METHOD AND MATERIALS

a: Processing pipeline overview.

A. Processing Pipeline Overview

The proposed method is an epoch-aware, double-thresholding pipeline that transforms a continuous multichannel EEG recording into channel-wise blink-region detections through five sequential operations. First, the EEG signals are downsampled and band-pass filtered before being divided into fixed-length, non-overlapping epochs. The proposed double-thresholding method then operates through three detection stages. Stage A performs multichannel peak-to-peak screening to identify candidate epochs containing prominent amplitude excursions. Stage B pools the samples from these candidate epochs to estimate a robust sample-level threshold independently for each channel. Stage C applies the resulting channel-specific thresholds to all valid epochs to localize blink regions and determine their epoch identifier, channel, onset time, and duration. Figure 1 summarizes the complete processing pipeline, while Stages A, B, and C are described in detail in the following subsections.

b: Preprocessing

First, the continuous recording is downsampled to a common sampling grid of 100 Hz. Standardising the sampling rate makes the amplitude thresholds and the minimum-duration gate comparable across recordings acquired on different hardware, and reduces the per-sample computational cost of the downstream stages without discarding any information needed to resolve a blink, whose spectral content lies well below the new Nyquist frequency.

Second, a zero-phase band-pass filter from 1.0 Hz to 20.0 Hz is applied which removes slow drifts and DC offsets at the low end and high-frequency muscle and line noise

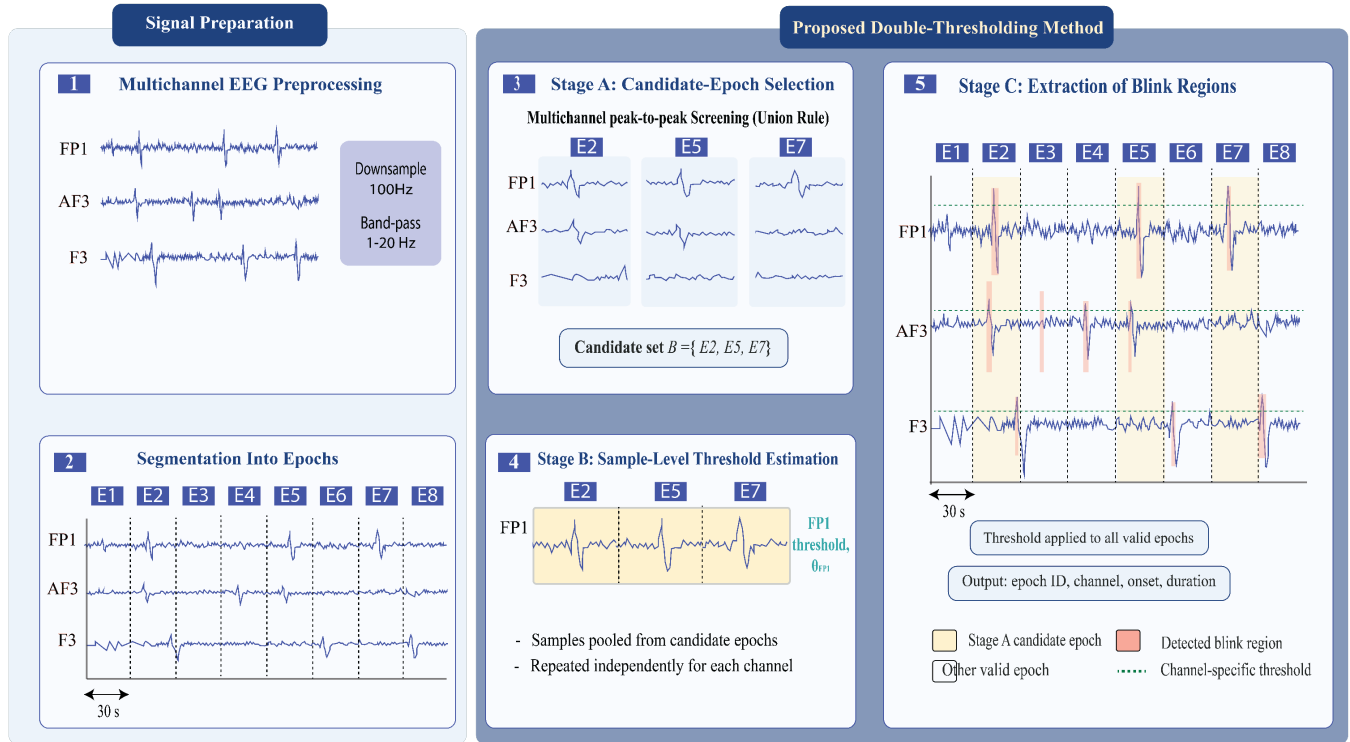


FIGURE 1. Overview of the proposed epoch-aware double-thresholding pipeline for EEG blink-region detection. Continuous multichannel EEG signals are downsampled to 100 Hz, band-pass filtered from 1 to 20 Hz, and divided into non-overlapping 30-s epochs. Stage A applies multichannel peak-to-peak screening and a union rule to select candidate epochs, represented by E2, E5, and E7 in this example. Stage B pools the samples from the selected epochs to estimate a robust sample-level threshold independently for each channel. Stage C applies the resulting channel-specific thresholds to all valid epochs and reports each detected blink region by its epoch identifier, channel, onset time, and duration. Pale yellow denotes Stage A candidate epochs, red denotes detected blink regions, and the dashed green lines represent the channel-specific thresholds.

at the high end, isolating the low-frequency, high-amplitude deflection that characterises an eye blink.

Third, each filtered recording is split into non-overlapping, fixed-length epochs. Epoching is what makes the thresholds *adaptive*, rather than deriving a single level for the entire session, the pipeline treats each short epoch as a local statistical context.

c: Selection of Candidate Epochs

The first stage, Stage A, of the proposed pipeline performs an initial epoch-level screening using an *autoreject*-inspired threshold-selection procedure to identify epoch that are most likely to contain prominent blink-related activity. Rather than relying on a single channel, the screening decision integrates channel-level information across multiple EOG/ EEG channels, allowing an epoch to be selected when abnormal amplitude excursions are detected in one or more channels. At this stage, the objective is not to localize individual blink samples, but to identify candidate epochs containing prominent amplitude activity. This multi-channel screening strategy therefore narrows the search space to potentially blink-related epochs for more detailed sample-level analysis in the subsequent detection stages.

Let $X \in \mathbb{R}^{N \times C \times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per

epoch. The sample at epoch $i \in \mathcal{E} = \{1, \dots, N\}$, channel $c \in \mathcal{C} = \{1, \dots, C\}$, and within-epoch time index $t \in \{1, \dots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1 \leq t \leq T} x_{i,c,t} - \min_{1 \leq t \leq T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c , a pattern consistent with ocular or other transient artifacts. Because signal amplitudes may vary across channels, participants, and recording conditions, a fixed peak-to-peak threshold is not imposed. Instead, a channel-specific threshold τ_c^* is calculated from each of the channel's. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion that measures the discrepancy between retained training epochs and validation epochs.

After the channel-specific thresholds, τ_c^* , have been estimated, the peak-to-peak amplitude of each epoch is evaluated separately for each EEG channel against its corresponding threshold. An epoch is retained as a candidate when the amplitude in at least one channel exceeds the threshold τ_c^* defined for that channel. Thus, the epoch-level screening decision incorporates information across multiple channels

using a union rule, rather than relying on a single predefined channel. The resulting set of candidate epochs is defined as $\mathcal{B}$. The set of suspicious epochs is therefore

$$\mathcal{B} = \{i \in \mathcal{E} : b_i = 1\}.$$

where b_i denotes the epoch-level screening outcome indicating that epoch (i) satisfies the selection criterion. These candidate epochs represent high-amplitude segments that may contain prominent blink-related activity and are retained for subsequent sample-level blink-region detection.

d: Computation of Sample-Level Blink-Region Threshold Candidates

Stage B estimates channel-specific sample-level amplitude thresholds using signal samples drawn from the candidate epochs $\mathcal{B}$ identified in Stage A. The estimation combines a measure of central tendency with a robust measure of dispersion and a sensitivity parameter, allowing the threshold to adapt to the amplitude characteristics of each channel while limiting the influence of isolated extreme samples.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity.

For each channel c , all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S}, 1 \leq t \leq T\}.$$

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \text{median}_{y \in \mathcal{Y}_c} y,$$

It should be note that, the arithmetic mean may be substituted when a mean-centered threshold is required for comparison.

The dispersion is estimated by the median absolute deviation (MAD),

$$\text{MAD}_c = \text{median}_{y \in \mathcal{Y}_c} |y - \text{median}_{u \in \mathcal{Y}_c} u|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\text{rob}} = 1.4826 \text{ MAD}_c.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD.

For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\text{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

A. EXTRACTION OF BLINK REGIONS

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^C$ obtained in Stage B to obtain channel-wise blink regions. Specifically, blink regions are searched for within each valid epoch $i \in \mathcal{E} = \{1, \dots, N\}$ and each channel $c \in \mathcal{C}$, by comparing the samples $x_{i,c,t}$ of that epoch and channel against the threshold candidate θ_c of the same channel.

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset time, and duration.

B. EXPERIMENTAL SETUP

The evaluation protocol was designed to compare all detector conditions on the same recordings, reference annotations, and scoring criteria. The datasets, event-level performance measures, and statistical analyses are described below.

1) Datasets

Two EEG corpora acquired during sustained simulated-driving tasks were used, providing 104 complete recording sessions in total. The corpora differed in their acquisition montages and data provenance, allowing the proposed method to be evaluated under different hardware setups.

The public Cao2018 corpus contains 62 sessions from 27 participants who performed an approximately 90-minute lane-keeping task in an immersive driving simulator [25]. The present study included the 58 sessions marked as complete and excluded the remaining four incomplete sessions. This study used the raw recordings. All 32 recorded electrophysiological channels (30 scalp electrodes and the A1/A2 mastoid reference channels) were supplied to the pipeline; the concurrently recorded vehicle-position channel was excluded. These 32 channels were Fp1, F3, Fp2, F4, FT7, FC3, C3, FC4, FT8, C4, TP7, CP3, TP8, CP4, T5, P3, O1, T6, P4, O2, F7, T3, Fz, FCz, Cz, Pz, Oz, F8, T4, CPz, A1, and A2.

The Internal corpus comprised 46 complete sessions recorded during a simulated lane-keeping task. EEG was acquired with a 128-channel EGI HydroCel system, with CPz as the reference. Although the original recordings used a high-density montage, the present analyses retained a predefined subset of 32 electrodes corresponding to conventional 10–20 scalp locations, following the subsampled 128-to-32-channel montage of [26]: Fp1, AF3, F3, Fp2, AF4, F4, FC5, FC1, C3, FC2, FC6, C4, CP5, CP1, CP2, CP6, P5, P3, PO3, O1, PO4, P4, P7, O2, F7, T7, Fz, Cz, Pz, Oz, F8, and T8.

We divided the 32 electrodes into four anatomical regions (frontal, central, parietal, and occipital) and further divided each region into left and right hemispheres, as per [27], as shown in Figure 2. In the figure, the resulting eight regions (frontal left, frontal right, central left, central right, parietal left, parietal right, occipital left, and occipital right) are abbreviated FL, FR, CL, CR, PL, PR, OL, and OR, respectively. The figure shows the scalp locations of the 32 electrodes: the blue circles represent electrodes located in both the Internal and Cao2018 montages, the yellow circles represent elec-

trodes located only in the Internal montage, and the red circles represent electrodes located only in the Cao2018 montage.

Of these eight regions, only the frontal, central, parietal, and occipital regions (left and right) were used in the analysis. The following electrodes were not assigned to any lateralized region and were likewise excluded from the analysis: F7, T7, F8, T8, Fz, FCz, Cz, CPz, Pz, and Oz.

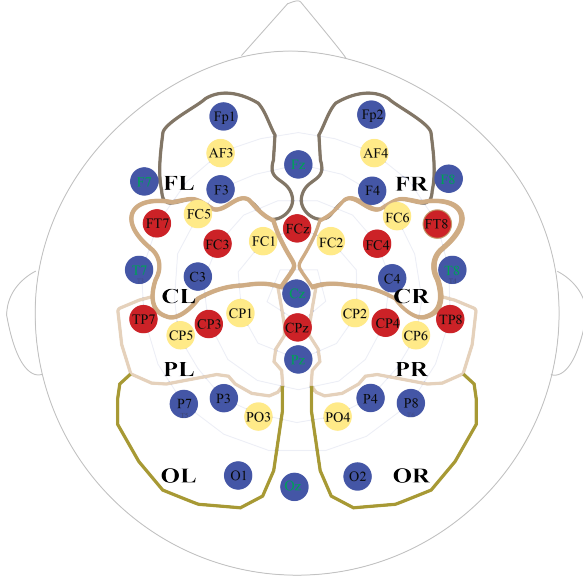


FIGURE 2. Scalp locations of the 32 electrodes used in this study, divided into eight anatomical regions (frontal left/right, central left/right, parietal left/right, and occipital left/right, abbreviated FL, FR, CL, CR, PL, PR, OL, and OR). Blue circles mark electrodes present in both the Internal and Cao2018 montages, yellow circles mark electrodes present only in the Internal montage, and red circles mark electrodes present only in the Cao2018 montage.

2) Evaluation Metrics

a: Event matching.

Performance was evaluated at the blink-event level rather than at the sample level. For a predicted interval P and a reference interval G , temporal overlap was quantified using intersection over union (IoU):

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \quad (1)$$

Here, $|\cdot|$ denotes interval duration. A predicted interval was eligible for a match when its IoU with an unmatched reference interval was at least 0.1. Matching was one-to-one, so a predicted or reference interval could contribute to at most one match. Matched predictions were counted as true positives (TP), remaining predictions as false positives (FP), and remaining reference intervals as false negatives (FN).

b: Session-level measures.

Precision, recall, and F_1 were calculated independently for each session s from the event counts:

$$\text{Precision}_s = \frac{\text{TP}_s}{\text{TP}_s + \text{FP}_s}, \quad (2)$$

$$\text{Recall}_s = \frac{\text{TP}_s}{\text{TP}_s + \text{FN}_s}, \quad (3)$$

$$F_{1,s} = \frac{2 \text{ Precision}_s \text{ Recall}_s}{\text{Precision}_s + \text{Recall}_s}. \quad (4)$$

Event-level F_1 was the primary outcome because blink events were sparse; sample-level accuracy would be dominated by non-blink samples and would therefore provide limited information about event-detection performance.

c: Macro-averaging.

For a measure $M \in \{\text{Precision}, \text{Recall}, F_1\}$, the reported macro-average over S sessions was

$$\bar{M} = \frac{1}{S} \sum_{s=1}^S M_s. \quad (5)$$

Thus, each session contributed equally, irrespective of its number of annotated blinks. Pooled results across the two corpora used the same unweighted average over all 104 sessions; event counts were not pooled before computing the measures. This approach prevents sessions with many blinks from dominating the headline result, as would occur with micro-averaging.

d: Operating-point selection.

Unless an analysis explicitly considered a fixed single electrode, each tested channel configuration was summarised at a retrospective *best-channel-per-session* operating point. Within each session, the output electrode with the highest event-level F_1 was selected, and its associated precision, recall, and F_1 were retained for macro-averaging. The same rule was applied to every detector condition in a comparison. Because this selection used the reference annotations, it represents an oracle operating point and should not be interpreted as prospective, label-free channel selection.

3) Statistical Analysis

a: Paired comparisons.

The recording session was the unit of analysis. Differences between detector conditions were assessed using the Wilcoxon signed-rank test applied to matched pairs of session-level F_1 scores. This non-parametric paired test was selected because F_1 is bounded and no normality assumption was imposed on the paired differences. All inferential tests reported in the main analyses were two-sided, and zero-valued paired differences were omitted according to the Wilcoxon convention.

b: Multiplicity and effect size.

The family-wise significance level was set to $\alpha = 0.05$. Bonferroni adjustment was applied separately within each pre-specified family of related comparisons. The four-condition strategy analysis comprised six pairwise tests, whereas the epoch-duration analysis comprised six comparisons between the non-reference durations and the 30 s reference. Channel-subset comparisons were corrected separately within each corpus over the 13 subsets tested against the full 32-electrode montage. Analysis-specific family sizes for any additional comparisons are stated with the corresponding results. Bonferroni-adjusted p -values are reported together with the matched-pairs rank-biserial correlation r as an effect-size measure; the sign of r indicates the direction of the paired difference in the stated comparison.

III. RESULTS

A. EXPERIMENT 1: EEG CHANNEL-SUBSET ANALYSIS

In the proposed pipeline, Stage A screens epochs using a union rule over the channels supplied to the pipeline, so changing the channel set can also change which epochs reach Stage B. Restricting the input channels therefore changes the detector itself, not merely the signals presented to a fixed detector. The EEG channel-subset analysis accounts for this coupling by re-running the complete pipeline separately for each channel subset. We first supply the complete 32-channel montage to the pipeline as a baseline, reported in Section III-A1. This configuration assumes no prior knowledge of which brain region projects blinks most strongly, making it the most general starting point against which reduced channel sets are compared. The analysis then evaluates anatomically defined regional subsets in Section III-A2 to identify where the EEG information most useful for blink detection is concentrated and to quantify the performance loss caused by excluding other scalp regions. Individual channels are subsequently assessed in Section III-A3 to determine whether a single optimally positioned electrode can preserve sufficient detection performance, thereby establishing the minimum channel configuration suitable for a simpler and more practical system.

1) Full-Montage Reference

a: Per-Electrode Detection

In this full-montage setup, per-electrode detection quality was concentrated at the frontopolar pair rather than distributed across the frontal region. On Internal, Fp2 and Fp1 reached F_1 values of 86.82% and 86.32%, respectively, before performance fell at the neighbouring frontal sites AF4 and AF3 to 80.32% and 73.76%, with F3, and F4 at 45.65%, and 45.23%. On Cao2018, Fp1 and Fp2 reached F_1 values of 79.68% and 78.45%, respectively, whereas the neighbouring frontal sites F3 and F4 were lower at 59.51% and 57.99%.

The central electrodes showed a dataset-specific asymmetry. On Internal, C3 reached $F_1 = 61.06\%$ while C4 reached $F_1 = 1.85\%$, whereas on Cao2018 C3 and C4 were close

at $F_1 = 31.08\%$ and $F_1 = 30.18\%$, respectively, so the pronounced C3–C4 asymmetry was specific to Internal.

The posterior electrodes performed markedly worse than the frontal and central sites, and the parietal and occipital groups also differed from each other between datasets. On Internal, the four parietal electrodes CP1, CP2, CP5, and CP6 were uniformly near floor, with F_1 confined to a narrow 1.75%–2.19% range, and the eight occipital electrodes P7, P3, PO3, O1, PO4, O2, P4, and P8 showed only slightly more spread, from $F_1 = 1.77\%$ at P4 to $F_1 = 3.60\%$ at PO3. On Cao2018, the two parietal electrodes were more informative but still well below the frontal and central regions, with CP3 reaching $F_1 = 17.84\%$ and CP4 reaching $F_1 = 19.56\%$. The Cao2018 occipital group was more heterogeneous than its Internal counterpart. O1 and O2 sat near floor at $F_1 = 3.10\%$ and 3.33%, T5 and T6 were intermediate at $F_1 = 9.26\%$ and 8.30%, and P3 and P4, the two most anterior electrodes assigned to this region, reached $F_1 = 12.80\%$ and 12.67%, respectively, so the region's mean F_1 was driven up by P3 and P4 rather than by the electrodes located over true occipital cortex.

2) Regional Channel Subsets

a: Per-Electrode Detection

Within these self-contained subset reruns, per-electrode detection quality remained concentrated at the frontopolar and neighbouring anterior frontal electrodes rather than distributed uniformly across the frontal region (Figure 3). On Internal, within the frontal subset's own rerun, Fp2, Fp1, AF4 and AF3 were the four strongest electrodes, reaching F_1 values of 85.28%, 84.84%, 77.04% and 70.70%, respectively. Performance then fell sharply at the two remaining frontal electrodes, F3 and F4, to 42.68% and 41.00%, respectively. On Cao2018, the same four strongest electrodes, Fp1, Fp2, F3 and F4, made up the entire frontal subset, reaching F_1 values of 78.80%, 77.05%, 55.97% and 54.64%, respectively. Although every one of these frontal electrodes scored lower in its own subset rerun than the same electrode did in the full-montage run (Section III-A1), the ranking of electrodes by F_1 was unchanged from the full-montage run in both datasets, so restricting Stage A/B screening to the frontal subset alone did not reorder the strongest frontal electrodes.

The central electrodes again showed a dataset-specific asymmetry within their own subset rerun. On Internal, C3 reached $F_1 = 58.59\%$ while C4 reached only $F_1 = 1.73\%$, so the pronounced C3–C4 asymmetry seen in the full-montage run persisted in the central subset's self-contained rerun. Whereas on Cao2018, C3 and C4 remained close together, at 27.67% and 26.62%, respectively, so this asymmetry stayed specific to Internal. Parietal and occipital electrodes were much lower in their own subset reruns. The best-performing parietal electrode was CP1 at 1.92% on Internal and CP4 at 15.63% on Cao2018, and the best-performing occipital electrode was PO3 at 3.35% on Internal and P3 at 10.04% on Cao2018. Overall, the self-contained subset reruns preserved the same anterior-to-posterior spatial pattern observed under

the full montage, with detection concentrated at the frontopolar and anterior frontal sites and much lower performance at the central, parietal and occipital electrodes.

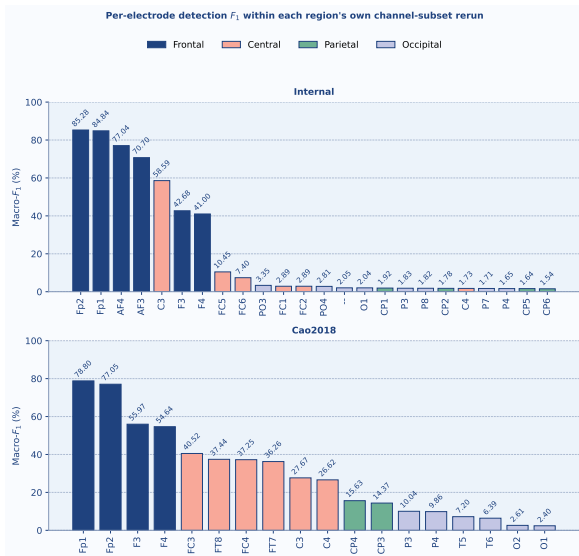


FIGURE 3. Per-electrode detection F_1 within each region's own channel-subset rerun, coloured by scalp region. Each bar is one electrode scored inside that region's own self-contained subset run.

b: Electrode-Level Significance Testing

The descriptive differences between the subset reruns and the full-montage reference were next tested at the individual-electrode level using session-paired comparisons. For every electrode entering one of the four anatomical subsets shown in Figure 3, that electrode's session-level F_1 inside its own subset rerun was paired against the session-level F_1 of the same electrode inside the full-montage run, on the sessions common to both runs. Each electrode was tested with a two-tailed Wilcoxon signed-rank test, paired by session, with p Bonferroni-corrected over every electrode tested within each dataset. Figure 4 shows, for every electrode, its session-mean F_1 under the subset rerun overlaid on the same electrode's session-mean F_1 under the full montage, together with the adjusted p -value.

On Internal (Figure 4A), 11 of the 24 electrodes tested remained significantly different after correction, and every significant difference was a reduction in the subset rerun. Despite a visible drop in the descriptive means, no statistically significant difference was detected for either of the two highest-performing electrodes, Fp2 and Fp1 (adjusted $p = 1.000$ for both). But, the remaining two of the four highest-performing electrodes, AF4 and AF3, did reach significance, with reductions that were statistically significant. Among the non-frontal electrodes, C3 showed a significant reduction whereas C4 did not, so the pronounced C3–C4 asymmetry in absolute F_1 persisted regardless of whether each electrode's own subset-versus-montage change was itself significant. Several parietal and occipital electrodes also

reached significance despite very low absolute F_1 (CP2, CP5, CP6, PO3 and PO4), while most others did not.

On Cao2018 (Figure 4B), significant differences were far more widespread. 16 of the 18 electrodes tested remained significant after correction, and again every significant difference was a reduction. As on Internal, the only electrodes not reaching significance were the frontopolar pair, Fp1 and Fp2 (adjusted $p = 1.000$ for both). The remaining two of the four highest-performing electrodes, F3 and F4, did reach significance. Unlike Internal, every central, parietal and occipital electrode tested on Cao2018 reached significance, including C3 and C4, whose reductions remained close together, consistent with the symmetry already noted between these two electrodes on this corpus.

Taken together, the electrode-level comparisons in Figure 4 show that a visibly lower descriptive mean did not by itself imply a statistically significant change. Fp1 and Fp2, the highest-performing electrode in each dataset, showed no detectable paired difference after correction in either corpus, while most other electrodes changed significantly, generally by a few percentage points. Significant differences were also more widespread on Cao2018 than on Internal, where roughly half of the non-frontal electrodes did not reach significance despite showing the same direction of change.



FIGURE 4. Per-electrode comparison of Proposed-Med F_1 when each electrode was evaluated inside its own self-contained anatomical subset rerun (narrow, opaque bar) against the same electrode evaluated inside the full-montage run (wide, translucent bar), for (A) Internal and (B) Cao2018. Bars show session-mean F_1 ; electrodes are grouped by the anatomical subset in Figure 3 and ordered by descending subset F_1 within each region. Comparisons are paired by session using two-tailed Wilcoxon signed-rank tests, with p Bonferroni-corrected separately within each dataset over every electrode tested there. An asterisk above an electrode marks adjusted $p < 0.05$; electrodes without an asterisk did not reach this threshold. Exact adjusted p -values are not printed on the bars and are instead reported in the text for electrodes discussed individually.

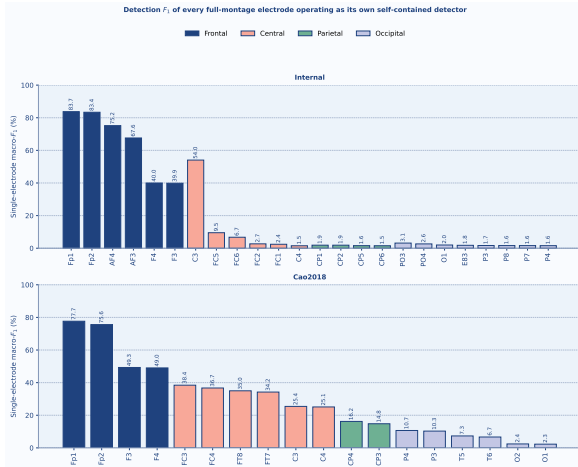


FIGURE 5. Detection F_1 of every electrode in the four curated anatomical regions operating as its own self-contained detector, in which the complete Stage A–Stage C pipeline was rerun on that one electrode alone, coloured by scalp region.

3) Single-Channel Operation

The regional channel-subset analysis above established how anatomical subsets of the montage behaved as self-contained detectors; the present analysis takes this reduction one step further and asks whether each *individual* electrode of the full montage can independently support the complete Proposed-Med detector. For every electrode retained in the full-montage configuration, the complete Stage A–Stage C pipeline was rerun using that one electrode alone, with the same 30-second epochs, median centre, preprocessing and thresholding parameters as the full-montage reference, so each result is a genuine self-contained single-electrode detector rather than one electrode’s score extracted from a multichannel run.

a: Per-Electrode Detection

Ranked by mean single-electrode macro F_1 , Fp1, Fp2, AF4 and AF3 topped Internal’s ranking, reaching F_1 values of 83.73%, 83.37%, 75.16% and 67.59%, respectively, the same four frontal electrodes identified by the earlier restricted analysis and in the same order except that Fp1 now leads Fp2 by 0.36 percentage points rather than trailing it (Figure 6; Figure 5). C3 followed next in the ranking at $F_1 = 54.05\%$, ahead of the remaining two canonical frontal electrodes, F4 and F3, which trailed further behind at 39.98% and 39.85%. Performance then fell sharply across the remaining 17 electrodes. The weakest was C4 at $F_1 = 1.51\%$, with every parietal and occipital electrode below 3.2%.

On Cao2018, Fp1, Fp2, F3 and F4 topped the single-electrode ranking, the same as Internal’s canonical frontal set, reaching F_1 values of 77.65%, 75.59%, 49.30% and 49.03%, respectively, matching both the identity and the order of the full-montage ranking (Figure 6). Performance then dropped substantially at the next electrode in the ranking, FC3, to $F_1 = 38.44\%$, followed by the remaining central electrodes FC4, FT8 and FT7, reaching F_1 values of 36.72%, 34.99%

and 34.22%, respectively. The central region therefore occupied the entire second tier of this ranking, clearly below the four frontal electrodes and still ahead of the weaker central pair, C3 and C4, at 25.43% and 25.10%. Beyond the central electrodes, performance declined further toward parietal (14.80–16.22%) and occipital sites, with the weakest electrode, O1, at $F_1 = 2.28\%$. This confirms, within the four curated regions, the same anterior-to-posterior decline reported for the full-montage and regional-subset analyses (Sections III-A1 and III-A2).

b: Electrode-Level Significance Testing

The descriptive differences between single-electrode and full-montage operation were next tested for every electrode using session-paired comparisons. For each of the electrodes, that electrode’s session-level F_1 operating alone was paired against the same electrode’s session-level F_1 scored inside the full-montage run, on the sessions common to both. For example, Fp2 operating as the entire detector was compared with Fp2 scored inside the full-montage Proposed-Med detector, and likewise for every other electrode. Each electrode was tested with a two-tailed Wilcoxon signed-rank test, paired by session, with p Bonferroni-corrected over the electrodes tested within each dataset. The Figure 6 shows, for every electrode, its session-mean F_1 operating alone overlaid on the same electrode’s session-mean F_1 under the full montage, together with the adjusted-significance marker.

On Internal (Figure 6A), 14 of the 24 electrodes tested remained significantly different after correction, and every significant difference was a reduction under single-electrode operation, with ΔF_1 ranging from -7.02 to -0.15 percentage among all 24 electrodes. Despite retaining the two highest absolute single-electrode scores, neither Fp1 nor Fp2 reached significance after correction (adjusted $p = 1.000$ for both), whereas the AF4 and AF3 did reach significance which is lower in the single-electrode relative to the full montage configuration. Beyond the frontal electrodes, the Internal C3–C4 asymmetry noted above coexisted with a significant configuration effect at both electrodes, and most occipital electrodes reached significance different except O1 and P4, despite all having very low absolute F_1 .

On Cao2018 (Figure 6B), 16 of the 18 electrodes tested remained significantly different after correction, again all reductions. Similarly, only the frontopolar pair, Fp1 and Fp2, did not reach significance after correction (adjusted $p = 1.000$ for both), and F3 and F4, the third- and fourth-highest single-electrode scores, showed the two largest absolute reductions ($\Delta F_1 = -10.21$ and -8.96 percentage points, respectively). Every central, parietal and occipital electrode reached significance, including both C3 and C4 (consistent with the C3/C4 symmetry already noted).

B. EXPERIMENT 2: STABILITY ACROSS EPOCH DURATIONS

Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink de-



FIGURE 6. Per-electrode comparison of Proposed-Med F_1 when each electrode in the four curated anatomical regions operated as a complete self-contained detector (narrow, opaque bar) against the same electrode scored inside the full 32-channel montage (wide, translucent bar), for (A) Internal and (B) Cao2018. Bars show session-mean F_1 ; electrodes are grouped by scalp region (frontopolar folded into Frontal) and ordered by descending single-electrode F_1 within each region, matching Figure 5. Comparisons are paired by session using two-tailed Wilcoxon signed-rank tests, with p Bonferroni-corrected separately within each dataset over the 24 (Internal) / 18 (Cao2018) electrodes tested there. An asterisk above an electrode marks adjusted $p < 0.05$; electrodes without an asterisk did not reach this threshold. Exact adjusted p -values are not printed on the bars and are instead reported in the text for electrodes discussed individually.

tector itself. In driving studies, epoch lengths ranging from a few seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F_1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

The pipeline was evaluated under epoch durations of 10, 20, 30 (reference), 40, 50, 60 and 120 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Experiment 1 found that Fp1 and Fp2 were consistently the two best-performing electrodes on both corpora. Each electrode is thus a deployable single-electrode operating point and the epoch duration is being varied. Figure 7 reports macro-averaged F_1 for each duration, with an asterisk marking the durations whose two-tailed Wilcoxon p -value, Bonferroni-corrected over the six non-reference durations within each electrode, differs significantly from the 30-second reference.

Internal showed a less consistent, and electrode-dependent, pattern. The 30 s reference yielded each electrode's highest Internal F_1 , 86.32% for Fp1 and 86.82% for Fp2. Relative to it, Fp1's F_1 fell significantly to 85.92% at 50 s ($p = 0.002$), to 85.67% at 60 s ($p < 0.001$) and to 85.04% at 120 s ($p < 0.001$). Fp2 differed significantly only at 120 s, falling to

85.57% ($p < 0.001$). Fp2's 50 s and 60 s declines, to 86.51% and 86.27% respectively, were descriptively similar to Fp1's but did not reach significance after correction, with $p = 0.080$ and $p = 0.064$ respectively. Neither electrode differed from its reference at 10, 20 or 40 s. Significance tracked the consistency of the change rather than its magnitude.

Performance on Cao2018 was stable for both electrodes, with no significant difference from the 30 s reference at any tested duration. For Fp1, the 30 s reference F_1 was 79.68%, the lowest value across the seven durations. At the two shorter durations, the F_1 score were higher than the 30 s reference value, reaching 79.81% at 10 s ($p = 1.000$) and 79.90% at 20 s ($p = 0.381$). The F_1 is in its highest value, 80.05% at 60 s, a peak 0.38 percentage points above the reference despite not reach significance ($p = 0.882$). For Fp2, the 30 s reference F_1 was 78.45%, also its lowest value. F_1 showed the same pattern, rising to 78.69% at 10 s ($p = 1.000$) and 78.71% at 20 s ($p = 0.159$), and reaching its highest value, 78.78% at 120 s, a peak 0.33 percentage points above the reference that also did not reach significance ($p = 1.000$).

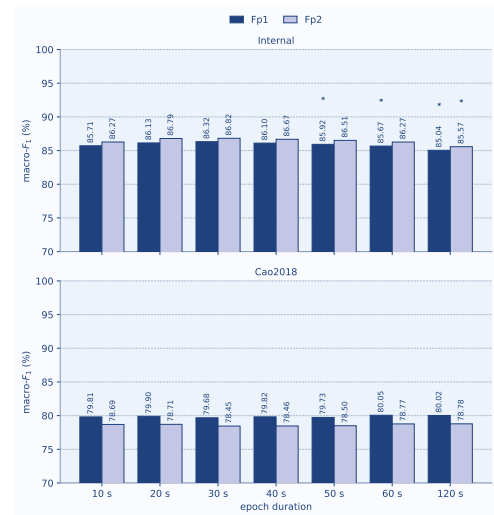


FIGURE 7. Macro-averaged F_1 of Proposed-Med across epoch durations, on the full 32-channel montage, for Fp1 and Fp2 (Experiment 1's two consistently best-performing electrodes) reported separately: Internal (top row) and Cao2018 (bottom row). Every value exceeds 75%, so the y-axis is clipped to 70–100% rather than starting at zero; each bar is labelled with its own value, and an asterisk marks the durations that differ significantly from the 30 s reference after Bonferroni correction.

C. EXPERIMENT 3: EFFECT OF THE THRESHOLD ESTIMATOR AT STAGE B

Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is selected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline such that it is contaminated by the very peaks the threshold is meant to capture. The mean and standard deviation are sensitive to these extreme values whereas the median and MAD are not. The hypothesis is that the proposed-Med will outperform Proposed-Mean on precision—because

the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall.

Within the two proposed configurations, the Stage-B threshold estimators were compared separately at Fp1 and at Fp2—the operating point Experiment 1 identified as consistently best-performing on both corpora (Table ??)—rather than a best-channel-per-session oracle, with each electrode's precision, recall and F_1 scored inside the full 32-channel montage run at the 30 s reference epoch (Table 1). At both electrodes and on both corpora, the direction of the metric changes did not support the stated precision hypothesis. On Internal, Proposed-Med achieved precision/recall/ F_1 of 86.45%/88.93%/86.32% at Fp1 and 86.28%/89.74%/86.82% at Fp2, compared with 87.04%/87.43%/85.55% (Fp1) and 86.75%/88.20%/85.99% (Fp2) for Proposed-Mean: F_1 rose by +0.76 (Fp1) and +0.83 (Fp2) percentage points, in both cases because a recall gain (+1.50 and +1.54 points) outweighed a precision loss (−0.59 and −0.48 points). On Cao2018, the same pattern held: Proposed-Med's F_1 rose by +0.29 (Fp1) and +0.43 (Fp2) percentage points, with recall gains of +1.96 and +2.14 points against precision losses of −1.29 and −1.28 points. Pooled performance followed suit, with F_1 increasing by +0.50 (Fp1) and +0.61 (Fp2) percentage points. None of the six dataset/electrode combinations reached significance under a two-tailed Wilcoxon signed-rank test on session-level F_1 (Internal: $p = 0.407$ at Fp1, $p = 0.435$ at Fp2; Cao2018: $p = 0.331$ at Fp1, $p = 0.795$ at Fp2; Pooled: $p = 0.791$ at Fp1, $p = 0.775$ at Fp2), so, as with the earlier best-channel-per-session comparison, the two estimators remain statistically indistinguishable at either fixed electrode. Pooled, Proposed-Med exceeded Proposed-Mean in 44/104 (Fp1) and 50/104 (Fp2) sessions, was equal in 2 and 4, respectively, and was lower in 58 and 50, indicating a per-session outcome close to even at both electrodes.

D. EXPERIMENT 4: STRATEGY COMPARISON

In driving and other epoch-structured paradigms, it is common to apply a continuous-signal blink detector to the raw or minimally preprocessed recording and subsequently map the detected blinks back onto the epoch grid. Both BLINKER and the standard MNE pipeline follow this approach, in which channel-specific thresholds are derived from a signal pool containing both blink-heavy and blink-free epochs. The central premise of the proposed pipeline is that pooling these heterogeneous epochs dilutes the amplitude statistics used for threshold estimation, thereby distorting threshold placement and shifting the resulting precision–recall trade-off.

Accordingly, we hypothesise that naive concatenation will produce systematically lower session-level F_1 scores than the epoch-aware pipeline because of a less favourable precision–recall trade-off. To evaluate this hypothesis, all four pipelines (i.e., BLINKER, MNE, Proposed-Med, and Proposed-Mean) are applied to every session using 30-second epochs and all available channels. All 32 channels are included to ensure a fair comparison with BLINKER, which, by default, processes all channels provided to it and requires the user to select a

TABLE 1. Effect of the Stage-B threshold estimator on the Internal and Cao2018 driving-EEG corpora at 30 s epochs, for Fp1 and Fp2 reported separately (not averaged). Proposed-Med (median/MAD) and Proposed-Mean (mean/SD) are each scored inside the full 32-channel montage run (“all_channel” gate); Fp1 and Fp2 are the two consistently best-performing electrodes identified in Experiment 1 (Table ??), so unlike a best-channel-per-session oracle, each is a fixed, deployable single-electrode operating point. Macro-averaged F_1 is reported as a percentage, per dataset and pooled over all sessions. Best F_1 per block in bold. p is a two-tailed Wilcoxon signed-rank test on session-level F_1 comparing Proposed-Med against Proposed-Mean within that electrode (one test per electrode; not Bonferroni-corrected).

Dataset	Channel	Condition	F_1 (%)
Internal	Fp1	Proposed-Mean	85.55
		Proposed-Med	86.32
		p vs. Proposed-Mean	0.407
Internal	Fp2	Proposed-Mean	85.99
		Proposed-Med	86.82
		p vs. Proposed-Mean	0.435
Cao2018	Fp1	Proposed-Mean	79.39
		Proposed-Med	79.68
		p vs. Proposed-Mean	0.331
Cao2018	Fp2	Proposed-Mean	78.01
		Proposed-Med	78.45
		p vs. Proposed-Mean	0.795
Pooled	Fp1	Proposed-Mean	82.12
		Proposed-Med	82.62
		p vs. Proposed-Mean	0.791
Pooled	Fp2	Proposed-Mean	81.54
		Proposed-Med	82.15
		p vs. Proposed-Mean	0.775

representative blink channel. For consistency across methods, all pipelines are therefore supplied with the same 32 channels, and the channel yielding the highest F_1 score is selected for subsequent analysis. The per-session macro- F_1 , precision, and recall are summarised in Table 2.

In the strategy comparison, both proposed configurations achieved higher macro- F_1 scores than BLINKER and MNE on Internal, Cao2018, and the pooled evaluation (Table 2). Proposed-Med achieved macro- F_1 scores of 88.25%, 80.68%, and 84.03%, respectively, whereas Proposed-Mean achieved 87.42%, 80.30%, and 83.45%. The corresponding scores were 76.54%, 69.40%, and 72.56% for BLINKER and 66.64%, 56.27%, and 60.86% for MNE.

This performance gap is reflected in the precision–recall profiles shown in Figure 8. BLINKER achieved the highest recall among all conditions on Internal, Cao2018, and the pooled evaluation, with values of 95.18%, 97.15%, and 96.27%, respectively, but also exhibited the lowest precision, at 67.94%, 56.96%, and 61.82%. Wilcoxon signed-rank tests on session-level F_1 , with Bonferroni correction across the six pairwise comparisons, showed significant differences between each proposed configuration and each baseline. Proposed-Med versus BLINKER yielded $p = 1.32 \times 10^{-11}$, and Proposed-Med versus MNE yielded $p = 6.51 \times 10^{-9}$. Similarly, Proposed-Mean versus BLINKER yielded $p = 4.97 \times 10^{-10}$, while Proposed-Mean versus MNE yielded $p = 9.94 \times 10^{-9}$. In contrast, neither within-pair comparison

was significant: BLINKER versus MNE yielded $p = 0.812$, whereas Proposed-Mean versus Proposed-Med yielded $p = 1$. Furthermore, no channel group in either dataset produced an inversion in which a baseline matched or exceeded the performance of Proposed-Med (Table 3).

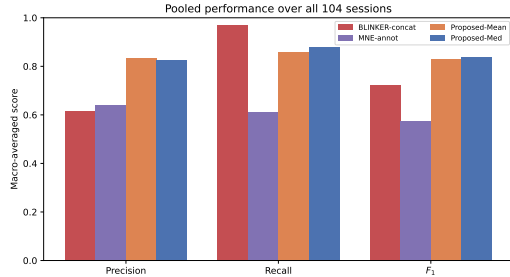


FIGURE 8. Macro-averaged precision, recall, and F_1 for the four conditions over all sessions.

E. OPERATING POINTS

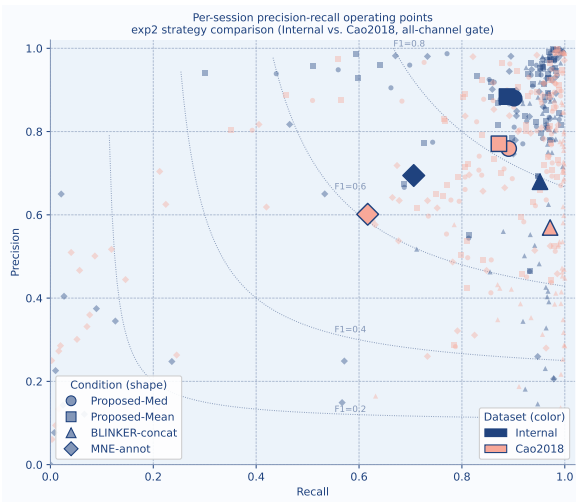


FIGURE 9. Per-session precision–recall operating points, Internal and Cao2018 shown separately.

Across the pooled and corpus-specific analyses, the precision–recall operating points showed a consistent ordering among the proposed and baseline conditions (Figure 9 and Figure 10). In the pooled analysis, Proposed-Med and Proposed-Mean occupied the high-precision, high-recall region, with precision and recall of 81.27% and 89.56% for Proposed-Med and 82.09% and 87.89% for Proposed-Mean, respectively. BLINKER-concat occupied the high-recall, low-precision region, with recall of 96.27% and precision of 61.82%, whereas MNE-annot was lower on both measures, with precision of 64.25% and recall of 65.64%. The same ordering was observed within each corpus. On Internal, the proposed conditions had precision of 88.03% and 88.40% and recall of 88.76% and 90.14%, whereas BLINKER-concat combined 95.18% recall with 67.94% precision and MNE-annot achieved 69.44% precision and 70.63% recall. On

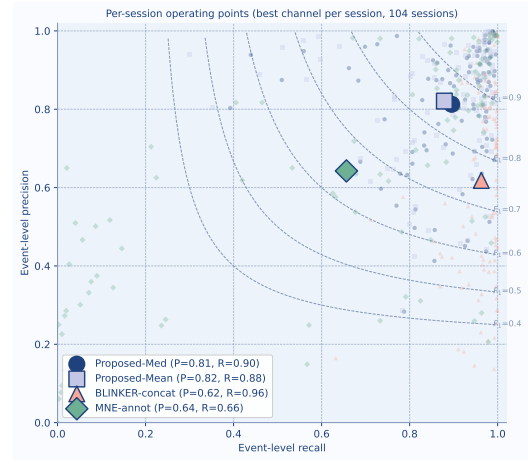


FIGURE 10. Per-session operating points pooled over both corpora, with condition means and iso- F_1 contours.

Cao2018, the corresponding pattern remained, with the proposed conditions retaining higher precision and recall than MNE-annot, while BLINKER-concat retained the highest recall and lower precision.

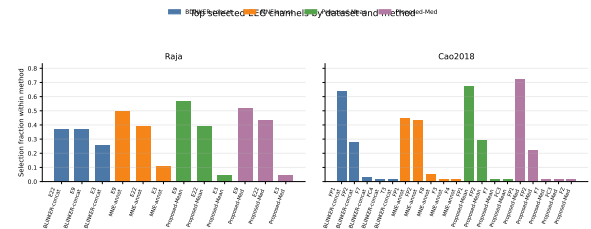


FIGURE 11. Best-channel selection frequency, pooled over the four conditions.

IV. DISCUSSION

Experiment 1 showed that useful detection information was concentrated at the frontopolar sites and that reduced frontal configurations retained substantially more performance than other anatomical subsets. Within the full-montage run, the best individual electrode was Fp2 on Internal ($F_1 = 86.82\%$) and Fp1 on Cao2018 ($F_1 = 79.68\%$). When the complete pipeline was instead run on one electrode alone, the best single-electrode result was obtained at Fp1 in both corpora, reaching 83.73% on Internal and 77.65% on Cao2018. Performance declined sharply for non-frontal subsets, although the subset analysis did not establish that every frontal channel was necessary. This concentration is compatible with the strong projection of blink activity to the prefrontal scalp and with earlier single-channel and forehead-EEG studies [10, 12, 15, 16, 21, 23, 28–30]. The findings therefore motivate further evaluation of reduced frontopolar systems for wearable or in-vehicle use, without establishing that such systems will reproduce the retrospective full-montage operating point in deployment.

The channel-choice analyses reported with Experiment 1 further showed that this spatial concentration did not translate

TABLE 2. Strategy comparison on the Internal and Cao2018 driving-EEG corpora at 30 s epochs. Each condition is summarised at its best-channel-per-session operating point (for every session the single channel or frontal sub-montage with the highest F_1 is selected, then averaged over sessions); the same rule is applied to all four conditions. Macro-averaged precision, recall and F_1 are reported as percentages, per dataset and pooled over all 104 sessions. Best F_1 per block in bold. By paired Wilcoxon signed-rank tests on session-level F_1 (Bonferroni-corrected over six pairs), Proposed-Med significantly exceeds BLINKER-concat ($p < 10^{-9}$) and MNE-annot ($p < 0.001$) but not Proposed-Mean ($p = 1.00$).

Dataset	Condition	Precision (%)	Recall (%)	F_1 (%)
Internal	BLINKER-concat	67.94	95.18	76.54
	MNE-annot	69.44	70.63	66.64
	Proposed-Mean	88.40	88.76	87.42
	Proposed-Med	88.03	90.14	88.25
Cao2018	BLINKER-concat	56.96	97.15	69.40
	MNE-annot	60.13	61.68	56.27
	Proposed-Mean	77.07	87.20	80.30
	Proposed-Med	75.91	89.09	80.68
Pooled	BLINKER-concat	61.82	96.27	72.56
	MNE-annot	64.25	65.64	60.86
	Proposed-Mean	82.09	87.89	83.45
	Proposed-Med	81.27	89.56	84.03

TABLE 3. Channel groups and datasets where a baseline algorithm equals or exceeds Proposed-Med in best-channel-per-session macro- F_1 . Values are percentages; F_1 is in percentage points.

Dataset	Selection	Baseline	BL- F_1 (%)	Prop- F_1 (%)	ΔF_1 (pp)
No inversions — Proposed-Med leads on every channel group.					

TABLE 4. Stability of the best-channel choice across the four conditions. Full agreement is the fraction of sessions where all four conditions select the same best channel; per-condition agreement is the mean fraction of the other three conditions selecting the same best channel.

Dataset	Sessions	Full agreement	Mean pairwise	BLINKER-concat	MNE-annot	Proposed-Mean	Proposed-Med
Internal	46	9/46 (0.20)	0.49	0.44	0.33	0.54	0.55
Cao2018	58	6/58 (0.10)	0.41	0.40	0.22	0.54	0.55
Pooled	104	15/104 (0.14)	0.44	0.42	0.27	0.54	0.55

into a stable choice of one electrode. Across the four conditions, Fp1 and Fp2 accounted for most selections: Internal selected Fp2 in 75/184 cases and Fp1 in 67/184, whereas Cao2018 selected Fp1 in 119/232 and Fp2 in 57/232. Nevertheless, all four conditions selected the same electrode in only 9/46 Internal sessions and 6/58 Cao2018 sessions, giving pooled full agreement in 15/104 sessions. The results therefore support stability at the level of the frontopolar region, but not interchangeability of Fp1 and Fp2 at the session level. Differences in ocular projection, electrode placement, and signal amplitude offer plausible explanations for this variability [24, 31], but these factors were not isolated in the present analysis. Moreover, because the reported best channel was selected retrospectively using reference F_1 , the results do not demonstrate a deployable adaptive channel-selection rule; they instead motivate evaluating a label-free rule for choosing between frontopolar channels.

Experiment 2 indicated that Proposed-Med, evaluated separately at Fp1 and at Fp2, the two electrodes identified in Experiment 1 as consistently the best-performing, was insensitive to epoch duration in the pooled analysis: macro- F_1

spanned only 0.41 (Fp1) and 0.50 (Fp2) percentage points across the seven tested durations, and none differed significantly from the 30-second reference after correction for either electrode (Figure 7). Cao2018 was similarly stable for both electrodes, with F_1 spreads of 0.38 (Fp1) and 0.33 (Fp2) percentage points and no significant difference at any duration. This pattern is useful because epoch duration is commonly determined by the experimental paradigm rather than by the blink detector. The corpus-specific results nevertheless qualify the pooled finding, and reveal an electrode-dependent asymmetry: on Internal, Fp1's F_1 declined from 86.32% at 30 seconds to 85.92% at 50 seconds ($p = 0.002$), 85.63% at 60 seconds ($p < 0.001$), and 85.04% at 120 seconds ($p < 0.001$), all three remaining significant after correction, whereas Fp2 declined significantly only at 120 seconds (86.82% to 85.57%, $p < 0.001$), with its 50- and 60-second drops falling short of significance ($p = 0.080$ and $p = 0.064$). Thus, the evidence supports stable pooled performance and no detected change from 10 to 40 seconds in either corpus for either electrode, but it does not establish equivalence across the complete 10–120-second range within every dataset, and the point at which Internal performance becomes unstable is itself electrode-dependent. Longer epochs may alter the composition of the signal used during screening and threshold estimation, but the present analysis did not test that mechanism directly.

Experiment 3 showed that the choice between the median/MAD and mean-based Stage-B estimators did not materially explain the performance of the proposed pipeline, evaluated separately at Fp1 and at Fp2, the two electrodes Experiment 1 identified as consistently best-performing, rather than at a best-channel-per-session oracle. The median/MAD centre was intended to reduce the influence of extreme values remaining in the suspicious-epoch pool, consistent with the robustness of median-based estimators to outlying observations [32, 33]. However, pooled precision fell from Proposed-Mean to Proposed-Med at both electrodes (80.49% to 79.51% at Fp1; 80.23% to 79.31% at Fp2), while recall rose (88.07%

to 89.83% at Fp1; 86.86% to 88.73% at Fp2) and F_1 increased modestly (82.12% to 82.62% at Fp1; 81.54% to 82.15% at Fp2). The estimators were statistically indistinguishable at either electrode (two-tailed Wilcoxon on session-level F_1 : $p = 0.791$ at Fp1, $p = 0.775$ at Fp2): pooled, Proposed-Med exceeded Proposed-Mean in 44/104 (Fp1) and 50/104 (Fp2) sessions, was equal in 2 and 4, respectively, and was lower in 58 and 50. The median centre therefore did not provide the hypothesised precision advantage and cannot be credited as a demonstrated source of improvement, at either electrode. It remains a defensible default on robustness grounds, but the measured comparison supports similar overall performance from the two estimators.

Experiment 4 showed that the epoch-aware proposed pipeline outperformed both comparison methods across the two driving corpora. Proposed-Med achieved a pooled macro- F_1 of 84.03%, compared with 72.56% for BLINKER-concat and 60.86% for MNE-annot (Table 2). BLINKER-concat represented an established blink-specific method applied to the concatenated recording [8], whereas MNE-annot represented the default amplitude-annotation routine from MNE-Python.

The corresponding session-level differences were statistically significant for both BLINKER-concat ($p = 1.32 \times 10^{-11}$) and MNE-annot ($p = 6.51 \times 10^{-9}$), and neither baseline equalled or exceeded Proposed-Med in any channel group on either dataset. These comparisons establish the advantage of the complete proposed configuration under the common evaluation protocol. Because the experiment compared complete detection strategies rather than a screening-on versus screening-off ablation of an otherwise identical detector, it does not by itself attribute the improvement exclusively to epoch screening.

The precision–recall profile is consistent with, but does not prove, the proposed explanation that heterogeneous epoch composition affects a threshold derived from the concatenated recording. BLINKER-concat had the highest pooled recall of 96.27% but substantially lower precision of 61.82%, whereas Proposed-Med combined recall of 89.56% with precision of 81.27%. BLINKER-concat therefore operated at a more permissive point, producing greater sensitivity together with more false detections. Variation in ocular-artifact amplitude across subjects, electrodes, and recording sessions makes sensitivity to the signal pool plausible [6, 8, 11, 24]. However, the reported experiments did not directly measure epoch composition, the resulting threshold shift, or the isolated effect of enabling and disabling epoch screening. The supported interpretation is consequently that the proposed pipeline achieved a more balanced operating point; a causal attribution to reduced threshold dilution requires a dedicated ablation.

These findings are subject to several limitations. The benchmark was restricted to two driving corpora, so generalisation to other experimental paradigms or clinical populations was not tested. Event detection was reported under one overlap-based matching rule, and sensitivity to alternative event-matching criteria was not evaluated; absolute F_1 values should therefore not be compared across studies without

accounting for their scoring rules [34]. Performance was not reported separately by blink duration, so detection of prolonged eye closures, which is particularly relevant to drowsiness assessment, remains unestablished. The evaluation also depends on the available reference annotations and their operational definitions of blink onset and offset, boundaries that can vary with signal representation and annotation criteria [1]. In addition, the headline strategy comparison used a retrospective best-channel-per-session rule based on reference F_1 ; this common oracle was applied fairly to every condition, but its scores do not represent prospective channel selection without labels. Finally, the proposed thresholding method was not benchmarked against learned detectors such as transformer-, vision-, or other deep-learning approaches [35–37]. Its demonstrated strengths are therefore limited to the methods and evaluation protocol examined here.

Future work should therefore evaluate prolonged eye closures separately, test prospective channel selection, and broaden the benchmark. Reporting normal blinks and prolonged closures independently would establish whether performance is retained for the sustained closures most relevant to drowsiness assessment [3, 5]. A prospective study should also replace the reference-based best-channel oracle with a fixed frontopolar electrode or a label-free quality criterion and quantify the resulting performance loss. Evaluation beyond driving data, including paradigms organised into task blocks, trials, or other natural segments, would test whether the observed robustness extends to different blink morphologies, cognitive demands, and recording conditions. Finally, the epoch-screening idea could be assessed as a front end to learned blink or drowsiness detectors [38–40]; such a hybrid remains a proposal and was not evaluated in the present study.

V. CONCLUSION

Taken together, the findings indicate that screening epoch-structured EEG before estimating sample-level thresholds provides more reliable event-level blink-region detection than deriving a single threshold from a concatenated recording under the evaluated driving conditions. The proposed median-based configuration outperformed the examined baselines across both driving corpora, supporting epoch screening as the principal methodological contribution of the pipeline. The spatial analysis further showed that frontopolar electrodes retained most of the full-montage performance, while the epoch-duration analysis indicated broadly stable pooled performance despite some corpus-specific sensitivity to longer epochs. Collectively, these findings support an interpretable and spatially sparse approach to epoch-aware blink detection that can accommodate common segmentation choices without requiring a dense electrode montage. The evaluation deliberately focused on driving data to establish the method's performance within its intended fatigue-driving context. Consequently, its generalizability beyond this domain and its suitability for practical deployment remain to be established. Future work should validate the pipeline across independent

driving datasets and recording systems before examining its transfer to other epoch-structured paradigms. Overall, this study positions epoch screening as a practical foundation for improving blink-detection threshold calibration in heterogeneous, epoch-structured EEG.

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